

Design of a Low-Cost Autonomous Epipelagic Profiling System for Oceanic Research

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Abstract— The epipelagic zone, the region from the surface of the ocean to 200 meters in depth, is an area of interest for ecologists, biologists, oceanographers, and other researchers interested in studying marine life and the environment. While autonomous underwater vehicles that can allow researchers to collect data throughout the epipelagic zone exist, most of these solutions are too expensive to enable smaller institutions and individual research labs to conduct their own explorations. This project aims to create an autonomous epipelagic zone profiling system that is relatively inexpensive (<\$2000), requires minimal maintenance, and is accessible to smaller and/or landlocked institutions. By providing a more accessible means of data collection, researchers can more cost efficiently conduct targeted studies of marine ecosystems to better understand the environment and topics such as the impacts of climate change on the oceans, or the changing population size of phytoplankton in the environment, etc. Through conversations with our stakeholders, Dr. Sheri Fløge of the Wake Forest University (WFU) Department of Biology and Electrical and Computer Engineer Dr. Kyle Luthy of the WFU Department of Engineering, we established that our system should be able to autonomously descend to a depth of 50 to 100 meters underwater and collect data such as water temperature, and pressure, as well as capture images. Using a systematic design process, the team was able to conceptualize a design for a low-cost modular buoyancy-controlled capsule. The capsule will be attached to a buoy system, to maintain its longitudinal and latitudinal position, from which it will be able to traverse the epipelagic zone to collect data. The team is currently in the process of prototyping and testing the system, and although the prototype will only have a few sensors, the modularity of the design will enable future users to purchase and attach various sensors (such as a PlanktoScope) that suit their needs. Over the coming weeks, the team will be completing assembly and conducting laboratory and field testing of the prototype.

Keywords— Autonomous; Submersible; Profiler; Data Collection; Oceanography

I. INTRODUCTION

The epipelagic zone is the entire area between the surface and 200m below the surface. Considering the oceans cover seventy-one percent of the earth's surface, the epipelagic zone represents one of the

largest habitats on earth. Scientists are interested in exploring this zone to investigate animal and plant life, climate change, and overall oceanic conditions. However, current available solutions for profiling this zone are prohibitively expensive and labor intensive, especially for smaller or inland institutions. The autonomous profiling float system will be designed and prototyped to provide an inexpensive means of exploration for environmental scientists, biologists, and other interested researchers to collect and retrieve data to a depth of 100m in the water column.

II. BACKGROUND

To better organize the scope of this project, the literature review process was broken down into three categories. These categories include research on the oceanic environment, challenges of operating a device in the ocean, and benchmarking current solutions.

- A. The pelagic region of the ocean has the largest volume of any habitat on earth, life in the pelagic zone is separated into three categories: the small phytoplankton, an important food source; the larger zooplankton, which subsist on the phytoplankton; and the nekton, the free-swimming organisms [1]. It is also separated into five zones: the epipelagic zone, from the surface to 200 meters; the mesopelagic zone, from 200 meters to 1,000 meters; the bathypelagic zone, from 1,000 to 4,000 meters; the abyssopelagic zone, from 4,000 to 6,000 meters; and finally the hadopelagic zone, from 6,000 to 11,000 meters [2]. Importantly in

only the epipelagic zone is there sufficient light for photosynthesis; therefore the team should only expect to collect meaningful data relating to photosynthesis in the first 200 meters of depth [2]. Based on these demarcations, this project will focus on the ability to profile about half of the epipelagic zone.

- B. Operating a device in the ocean poses unique challenges, specifically the effects of corrosion, pressure, and biofouling. From sciencing.com the team learned that when metal is submerged in saltwater it corrodes [3]. From the National Oceanic and Atmospheric Administration, the team learned that hydrostatic pressure increases with depth, specifically “for every 33 feet (10.06 meters) you go down, the pressure increases by one atmosphere.” [4] Pressure considerations will be central to designing the structure of the device. According to the International Maritime Organization, biofouling refers to “the undesirable accumulation of microorganisms, plants, algae and animals on submerged structures.” [5]
- C. For benchmarking, existing solutions that are currently available were explored. Specifically, the BGC-ARGO, a network of profiling floats that collect data for biological, geological, and chemical research. The BGC-ARGO floats measure oxygen concentration, nitrate concentration, pH, chlorophyll *a* fluorescence, suspended particles, and downwelling irradiance [6]. This list of sensing capabilities can be used to compare this device’s abilities to the team’s device. Using these projects, the team is able to study the design decisions that other researchers have made. For example, the buoyancy system on the BGC-ARGO was used to inspire the buoyancy control concept.

III. SYSTEM REQUIREMENTS

The system requirements were derived primarily through the provided problem description, interviews with stakeholders, research, and team discussion. A list of desired sensors and functions was received

from one of the primary stakeholders, Dr. Floge, an expert in the field of oceanography that is currently conducting research in this area. Along with this, conversations with Dr. Luthy, an expert in the field of electrical and computer engineering, specifically in aquatic vehicles, also defined some system requirements. The compiled list of system requirements is included below along with their weight of importance on a scale of 1 (low importance) to 10 (high importance).

TABLE I
SYSTEM REQUIREMENTS

Weight of Importance	System Requirements
10	Autonomously navigate through the water column (0 to 100m in depth)
8	Be contained within commercially available watertight enclosure
8	Only include sensors that can withstand pressure at 100m (10bar)
6	Include an automated “Abort System” that returns device to the surface in the case of flooding
6	Record an accurate timestamp for each measurement taken
6	Collect and store still images of the surrounding ocean
6	Measure the water pressure exerted on the vehicle to an accuracy of 1.25 psi
5	Be able to be constructed within the \$1500 budget
4	Measure the temperature of the surrounding water to an accuracy of +/- 0.1 degree

IV. CONCEPT DEVELOPMENT/DESIGN

Concept generation was divided into two main spheres. The first was component exploration, or market research, and the second was concept generation for the device's propulsion system which represents the majority of the project's mechanical challenge. A component search and comparison was conducted to identify components that could satisfy the identified system requirements. These components included sensors, electronics, structural components, etc. As for the propulsion system, four

concepts were generated: propellers, wave ratcheting, water intake, and variable buoyancy propulsion. These concepts were generated through communication with the lead stakeholder Dr. Luthy, as well as through team discussion, background research and benchmarking. Utilizing a weighted decision matrix and a Pugh chart, the variable buoyancy propulsion concept was selected. The system would be completely sealed from the outside. An internal ballast is filled with oil, and when the system needs to be positively buoyant a motorized mechanism will pump the oil into an external bladder, inflating it. This increases the total volume displaced and therefore decreases the density, allowing it to float. When it needs to be negatively buoyant, the ballast draws the oil back in from the external bladder and decreases the displaced volume. The system was chosen because of its power efficiency compared to propellers, and its superior degree of controllability compared to a wave ratchet system. It also beat out the water intake concept for its higher reliability from being a closed system. Finally, the system has the best design flexibility as it can be switched to a water intake design with a minimal amount of adjustments.

V. EMBODIMENT DESIGN

From the system requirements it was determined that four sensors were needed for this prototype, these features include a pressure sensor, temperature sensor, leak sensor, and high quality camera.

All of the internal components are housed inside two 4 inch diameter cast acrylic tubes that are 12 inches in length, purchased from Blue Robotics, both rated for 100m depth [7]. The tubes are capped with aluminum end caps and O rings to prevent leaks. The two capsules are individually sealed and connected by a pass through cable between them. A custom brace has been designed to hold the two capsules together, the upper capsule houses the buoyancy engine, and the bottom capsule houses the electronics, battery, and sensors. The bottom capsule will have a clear acrylic dome on the end to allow the camera to capture images facing downwards. This modular design is modeled to be expandable, and

since each capsule is individually sealed, if water breached one of the capsules the other one will be safe. Fig. 1 depicts the general structure of the system design.

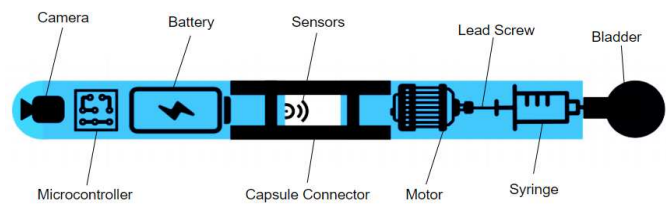


Fig. 1 Design Layout

To hold all the electronics in place, two custom electronics trays were designed to maximise space while securing all the components (Fig. 2). Along with this, a modified end cap was produced with a hole that allows for the external ballast to be threaded through.



Fig. 2 Buoyancy Engine Tray (Left) and Sensor Tray (Right)

The buoyancy engine is designed around inflating and deflating an external ballast to change the displacement of the system. Based on the calculations in Fig. 3 it was determined that the variable displacement of the ballast should be around 250 cm³ to accommodate the total displacement.

4942cm^3 [base minimum displacement] $\times (1.05\text{g}/\text{cm}^3)$ [target density for negative buoyancy in seawater] = 5200g [target weight of system]

$5200\text{g} = 5200\text{cm}^3$ [target density for positive buoyancy in seawater]

$$5200\text{cm}^3 - 4942\text{cm}^3 = 258\text{cm}^3 \text{ [variable displacement]}$$

Fig. 3 Engineering Calculations for the Buoyancy Engine

A modified syringe is designed to inflate and deflate the ballast. The syringe will be powered by a stepper motor connected to a lead screw for linear actuations, retracting and extending the syringe plunger. The syringe is connected to the ballast via a rubber tube and will inflate and deflate the bladder when it is actuated, as shown in Fig. 4.

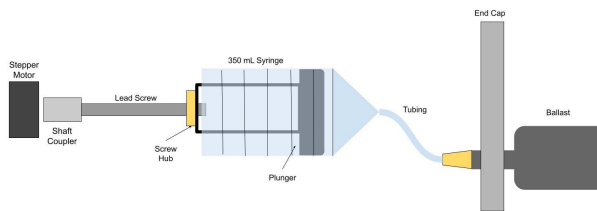


Fig. 4 Buoyancy Engine Diagram

In terms of controls, two concepts were generated. The first is using an Arduino. The second concept is to use a Sleepy Pi 2 to control the buoyancy engine; it would be connected to a Raspberry Pi, which handles the sensing functions. The Sleepy Pi 2 turns on the Raspberry Pi every time a target depth is reached and turns it back off when cruising to maximise battery life. This way the Raspberry Pi can stay dormant when the sensor functions are not needed. The Sleepy Pi 2 and Raspberry Pi combination was chosen because of the superior computational power of the Raspberry Pi, which allowed for the capturing of images and local storage of data. Also the Raspberry Pi 4's capabilities as a processor lends itself to the possibility of expanding the systems sensing capability at a later date.

The operating process of the system is separated into three steps. First, the system deflates the ballast and descends in the water column. Then, when the system reaches a target depth, the pressure sensor

triggers and the system inflates the ballast to half full and becomes neutrally buoyant to stay in the target depth. Lastly, the Sleepy Pi 2 activates the Raspberry Pi and the system begins to collect data using the sensors and camera. After the process is complete, the Raspberry Pi shuts off and the process repeats itself from step one. When the system reaches its maximum depth, it ascends with the same procedure.

The completed design assembly was modeled in CAD (Fig. 5) using Autodesk Fusion 360 and updated throughout the process to assist the team with prototyping and testing. Individual components were first placed in the CAD model to determine spacing and fit. The electronics trays were designed around the complete model to maximize space efficiency. The model was also used to demonstrate the proof of concept prototype during design reviews and presentations.

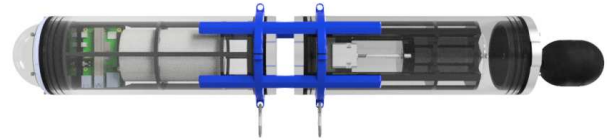


Fig. 5 Complete Assembly (CAD Model)

VI. TESTING AND REFINEMENT

Significant testing was performed to evaluate the prototype. First, the custom designed electronics trays were inserted and integrated with all of the electronics and buoyancy engine. Multiple iterations of the trays were rapidly prototyped using 3d printing. Each iteration was designed to be more space efficient and have more structural integrity.

The connecting brace was first prototyped with PLA plastic. Subsequent iterations improved the structural rigidity and the material was changed to ABS plastic (Fig. 6) to avoid salt water erosion and increase the longevity of the system while in situ.



Fig. 6 Connecting Brace

On the controls side, the team started by testing the sensors and motor with an Arduino as the microcontroller. Once everything was working as intended, the team switched over to the Sleepy Pi 2 and Raspberry Pi combination.

The buoyancy engine was constructed out of a commercially available 500ml syringe that was modified to hold approximately 250ml. The stock plunger was replaced with a 3d printed plunger shown in Fig. 7 for better structural rigidity and interfacing with the lead screw assembly.



Fig. 7 Custom Plunger

To connect the buoyancy engine to the external ballast, a custom fitting needed to be made. An aluminum end-cap was machined with a waterjet, drilled, and tapped to fit the exact dimensions of the external ballast. The ballast is then threaded through the end-cap and the connection is potted with marine epoxy. The end-cap went through two iterations so that space within the capsule could be maximized to

accommodate the other propulsion system components.

VII. CONCLUSIONS

This system was developed through a process of concept generation, evaluation of existing solutions, background research, and iterative prototyping. Based on the system requirements derived from the problem description, conversations with clients, and team discussions variable buoyancy propulsion was determined to be the best fit for the system. To implement this propulsion method, an internal ballast filled with oil is controlled by a motor to pump oil into an external ballast which increases volume and decreases density, allowing the system to be positively buoyant. For the system to be negatively buoyant, the motor controls the internal ballast to draw oil from the external ballast which has an inverse effect.

The hull of the system consists of two capsules connected by a custom brace, one capsule is sealed by a modified end cap that is attached to the external ballast. Two custom electronics trays are used to hold the inner components, including the motor, battery, and controls. The sensor capabilities of the system include pressure, temperature, and leak sensors as well as a camera for optical sensing. Though the system will not be equipped with many sensors, it is designed with modularity in mind and will therefore allow the addition of more capabilities in the future. The fully assembled system is shown in Fig. 8.

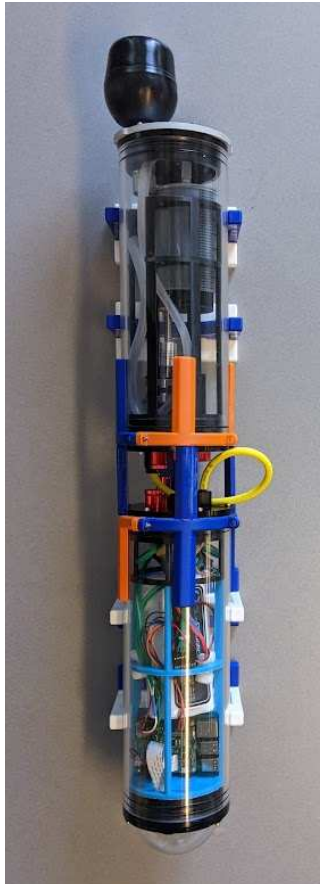


Fig. 8 Complete System Assembly

Though the system is near completion further testing is necessary. The first test is the watertightness of the capsule with the machined end cap. To ensure the reliability of the end cap, a vacuum pump will be used to determine air-tightness, and to determine if the end cap is well fitted. Finally, the propulsion system will need to be tested; to do this, the system will be tethered and monitored in a deep pool, the system will increase and decrease its buoyancy to test its functionality.

VIII. FUTURE WORK

Future work for this system will include fine tuning the propulsion control code in order to allow users to dictate the desired depths of interest. Control theory methods such as PIDs will be implemented and tuned to maximize efficiency and accuracy. The system will also need to be tested in saltwater and go through

endurance testing to make sure the exterior can withstand the saltwater corrosion. This system was designed with specific clients in mind; Dr. Luthy and Dr. Flöge will be using this system to profile the epipelagic zone off the coast of North Carolina. The final iteration of our design will include depth, temperature, leak, and optical sensing. Dr. Luthy and Dr. Flöge plan to add a PlanktoScope, a compact digital microscope; as well as a CTD (conductivity, temperature, and depth) sensor, which will allow for more precise results. Further add-ons are also possible through the modular aspect of the design by adding more capsules to the system. The Raspberry Pi unit should also contribute to the expandability with its robust computational capabilities.

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